

1.17 EXERCISE

1. How do you define a robot? What is robotics?
2. Is CNC machine a robot? Explain it.
3. What do you mean by dof of a system? Why do we generally use 6 dof manipulators for industrial purposes?
4. What is a redundant manipulator? When and why do we need it?
5. How do you classify the robots based on the coordinate system?
6. Dextrous workspace of a manipulator is the sub-set of its reachable workspace – justify the statement.
7. Define the terms: resolution, accuracy and repeatability of a robot.
8. How do you specify a robot?
9. Prepare a detailed report on applications of robots in various fields.
10. State the differences between: (a) internal gripper and external gripper, (b) active gripper and passive gripper, (c) hard gripper and soft gripper.
11. Explain the working principle of (a) magnetic gripper, (b) vacuum gripper.
12. Explain 'peg into hole' problem. How does a remote center compliance solve this problem?
13. Fig. 1.22 shows the schematic view of a mechanism. Determine its dof. Make comments on the result.
14. Fig. 1.23 shows the schematic view of a slider-crank mechanism. Determine its dof. Make comments on the result.

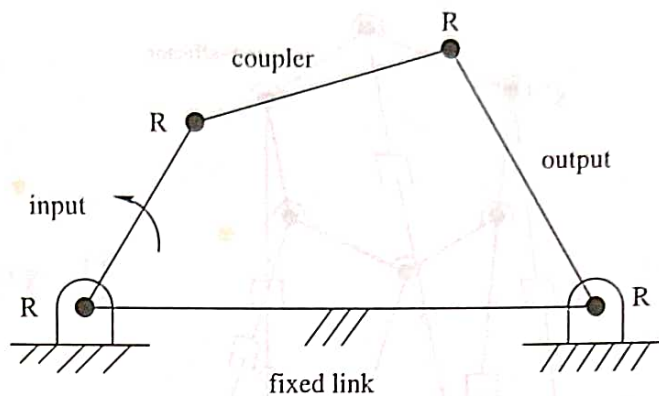


Figure 1.22: A four-bar mechanism.

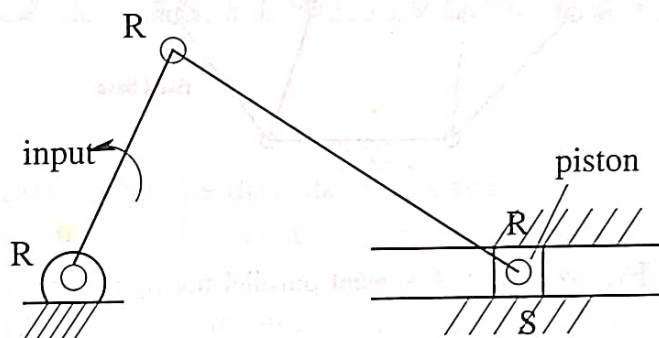


Figure 1.23: A slider-crank mechanism.

15. Draw kinematic diagram of a human hand. Determine its dof.
16. Let us suppose that a manipulator is going to clean a vertical black-board with the help of a duster. What is the minimum dof the manipulator should have? Explain.
17. What do you mean by on-line and off-line teaching methods of a robot? Explain the methods of manual teaching and lead-through teaching methods used in robots.
18. Fig. 1.24 shows a spatial parallel manipulator. Determine its dof.

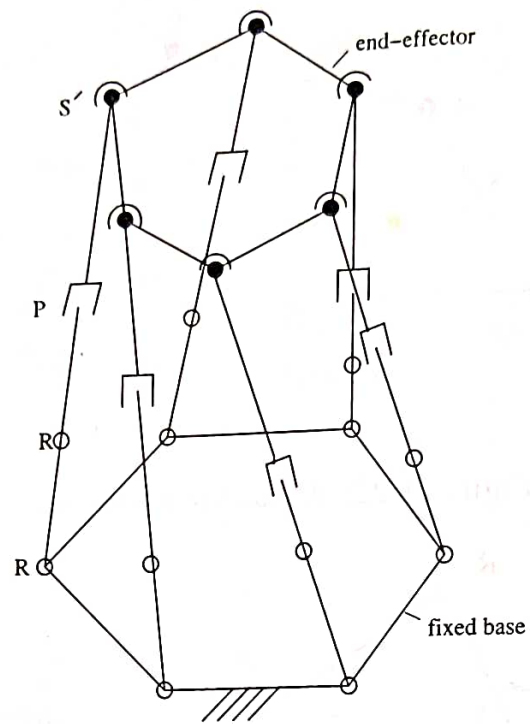


Figure 1.24: A spatial parallel manipulator.

19. Draw the workspace of a latch-bolt system shown in Fig. 1.25.

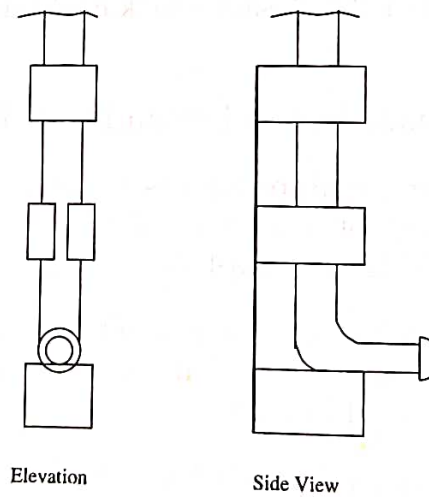


Figure 1.25: A latch-bolt system.